

KVCache Practices at MiniMax for Agentic Workloads

From Traffic Characteristics to Architectural Insights

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MiniMax: Every Modality. Every Model. All Apps. Global SOTA



All-Modality SOTA: The Entry Ticket to the Final Table of Global AI Leadership

Applications



Agent



Hailuo AI



MiniMax Audio



Talkie



Open Platform

Models

LLM: M2.5 & Agent

- **SOTA Benchmarks:** SWE-Bench Verified 80.2% (Coding); BrowseComp 76.3% (Search); BFCL 76.8% (Tool-calling)
- Optimized for efficient execution, **37% faster** at complex tasks.
- 10B activation and custom chips
- **\$1/hour with 100 tps**

Speech-2.8

- **<250ms Ultra-Low Latency** — Real-time interactive use cases
- **High-Fidelity Voice Cloning** for 40+ Languages

AI RACE FINAL TABLE

All-Modality Frontier Models



OpenAI



MINIMAX



Google



ByteDance



Alibaba.com®



deepseek

ANTHROPIC

Meta

Tencent

Video: Hailuo-2.3

- **World-Class Quality and Record-Breaking Cost Efficiency**
- **Handles Extreme Physics** — Outstanding actions and micro-expressions
- **Hyper-Realistic Motion**

Music-2.5

- **Conductor-Level Precision** — 100+ instruments at studio-grade quality
- **14+ Structural Tags** — Full-track control with genre-specific design

Infra

Large-scale Training and Inference Platform

MiniMax: One of the Largest Global Token Exporters



Global Leader in All Modalities

First-tier performance across global benchmarks in text, video, and audio models



4.5tn+

Average daily text tokens processed in March 2025



1.6mn+

Average daily video generated in March 2025



370k+ hours

Average daily audio generated in March 2025



200+

Coverage of countries and regions



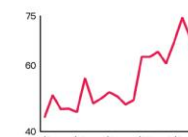
200mn+

Users served globally



50,000+

Cumulative enterprises and developers served globally



+70%

Stock Price Return in 27 days; \$20B mkt cap

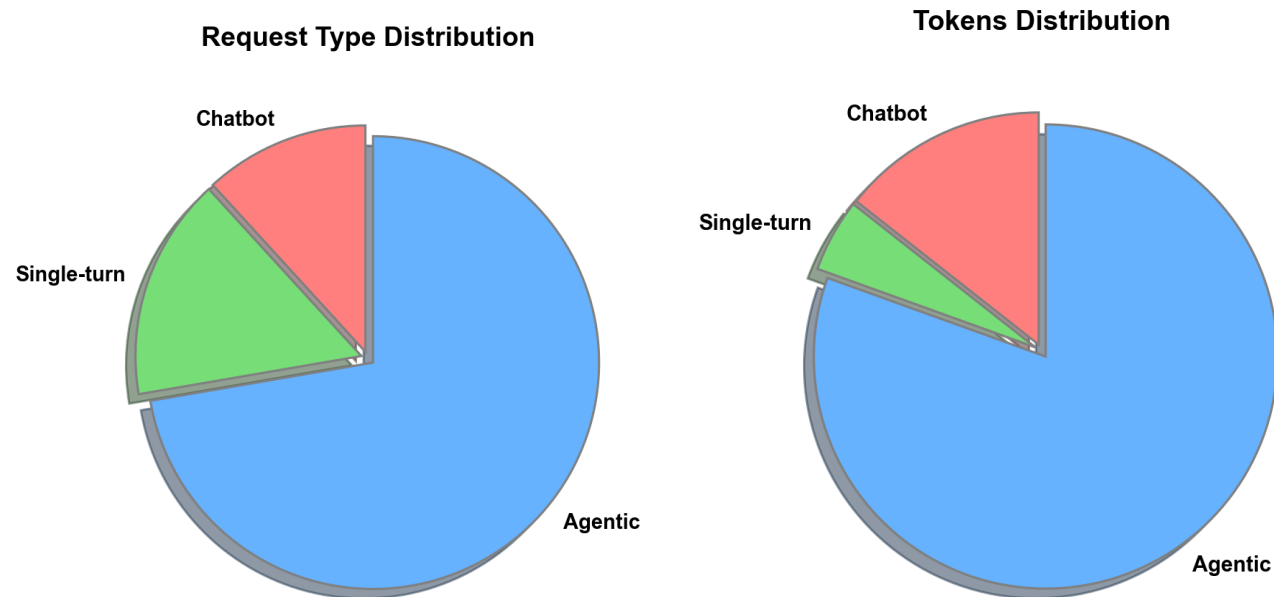
Overview

- Part 1: Agentic Workload Pattern
- Part 2: KVCache Management at MiniMax
- Part 3: Thoughts about Evolution of KVCache Management

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Workload Overview

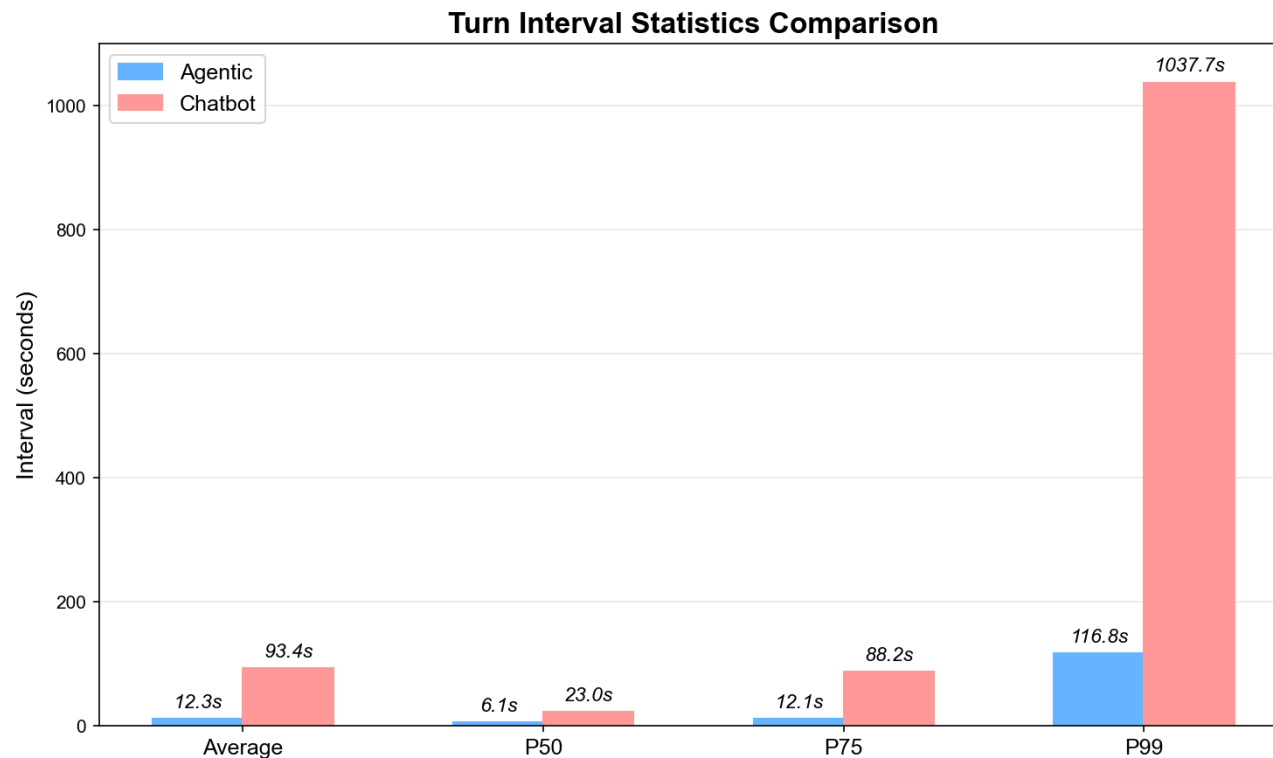
- Typical workloads: chatbot, agentic
 - Agentic is the majority!
- Temporal Locality
 - Time interval between requests
- Spatial Locality
 - Reuse ratio per request
 - Reuse times
 - Incremental prefill tokens(uncached tokens)



	Time Interval(Temporal)	Reuse Ratio(Spatial)
chatbot	long, human-in-the-loop	mostly high, except long document processing
agentic	very short, without human involved	very high in single request, and hit more times

Agentic Workload

- Temporal Locality
 - p50 time interval: 6s (kvcache write/read span)
- Spatial Locality
 - p50 request length: ~60k tokens
 - p50 chat rounds: ~10 rounds
 - theoretical reuse ratio: ~90%
 - actual reuse ratio: ~80%
 - p50 incremental tokens: ~740 tokens
 - size per tokens in M2.5(8bit): ~130KiB/token



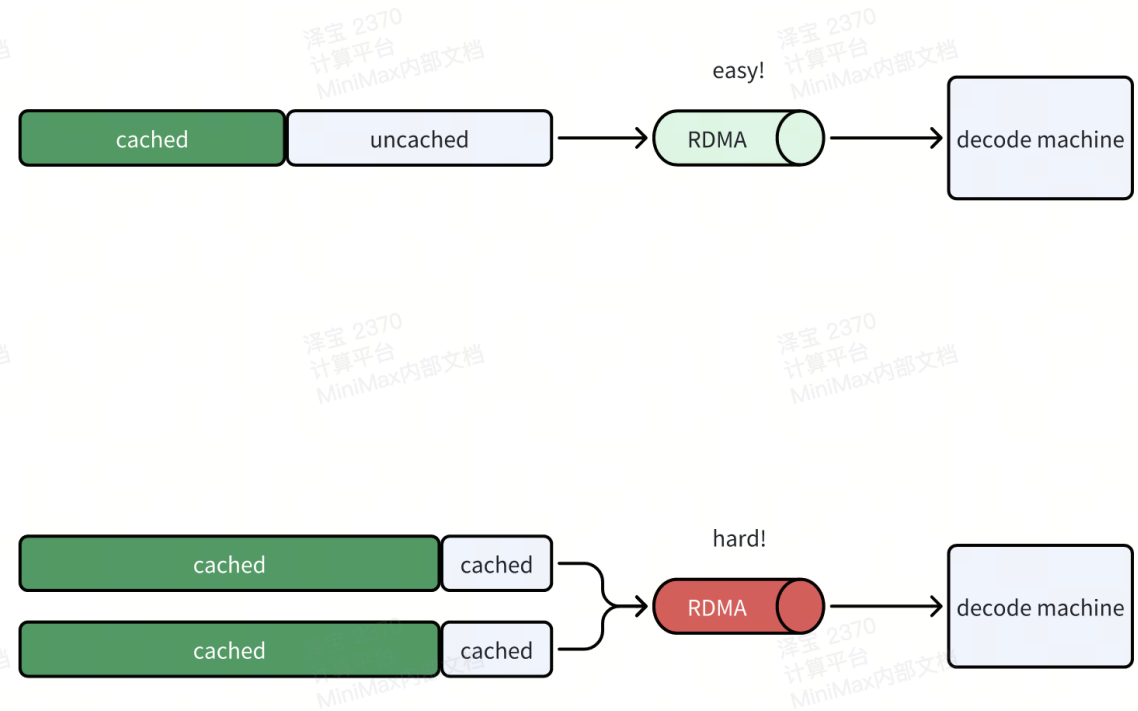
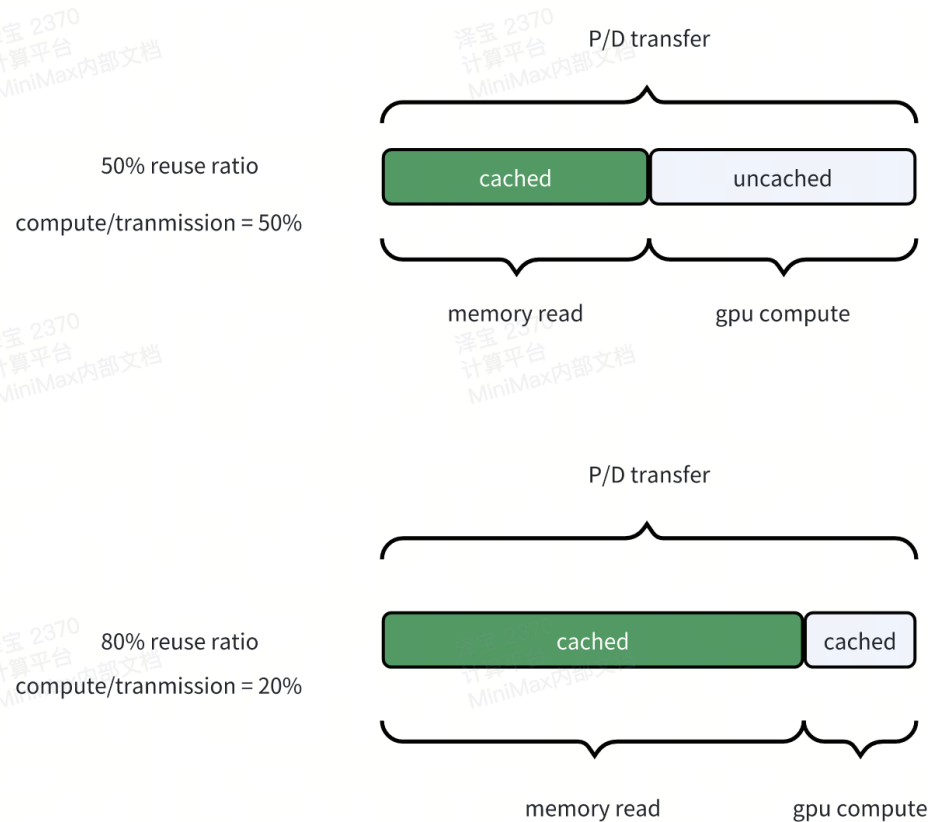
Agentic Patterns: Challenges

- LLM serving performance goal:
 - SLO: TTFT, TBT
 - Resource usage: GPU
- Challenge 1: High transmission pressure
 - less uncached tokens, higher transmission/compute ratio
- Challenge 2: High storage bandwidth pressure
 - high reuse ratio & short interval causes high storage bandwidth pressure, low effective capacity
- Challenge 3: Sensitive to cache miss & QPS burst
 - high reuse ratio causes high penalty of cache miss and sensitivity to QPS burst

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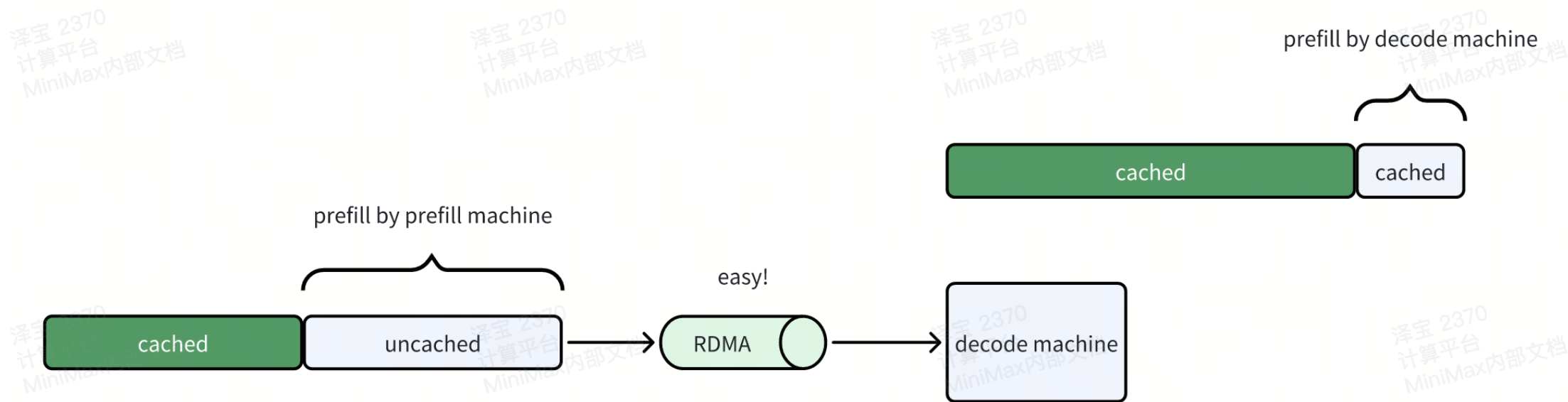
Challenges & Solutions

- Challenge 1: High transmission pressure
 - less uncached tokens
 - higher transmission/compute ratio
 - GPU waiting, low usage



Challenges & Solutions

- Solution 1: Decode can prefill a little
 - reduce transmission by decode doing prefill
 - only when incremental prefill tokens < threshold



Challenges & Solutions

- Challenge 2: High storage bandwidth pressure
 - high reuse ratio, short interval
 - bandwidth demand = (reused kvcache size per request) * (QPS per node) / (kvcache load time SLO)
 - ~7GiB KVCache(M2.5, 8bit kvcache) to load
 - SLO: TTFT < 1s, KVCache load time < 500ms
 - low capacity demand = (storage bandwidth * interval)
 - high bandwidth/capacity ratio demand: SSD(1000)
 - high tail latency
- Solutions:
 - estimate load time
 - prefetch to overlap

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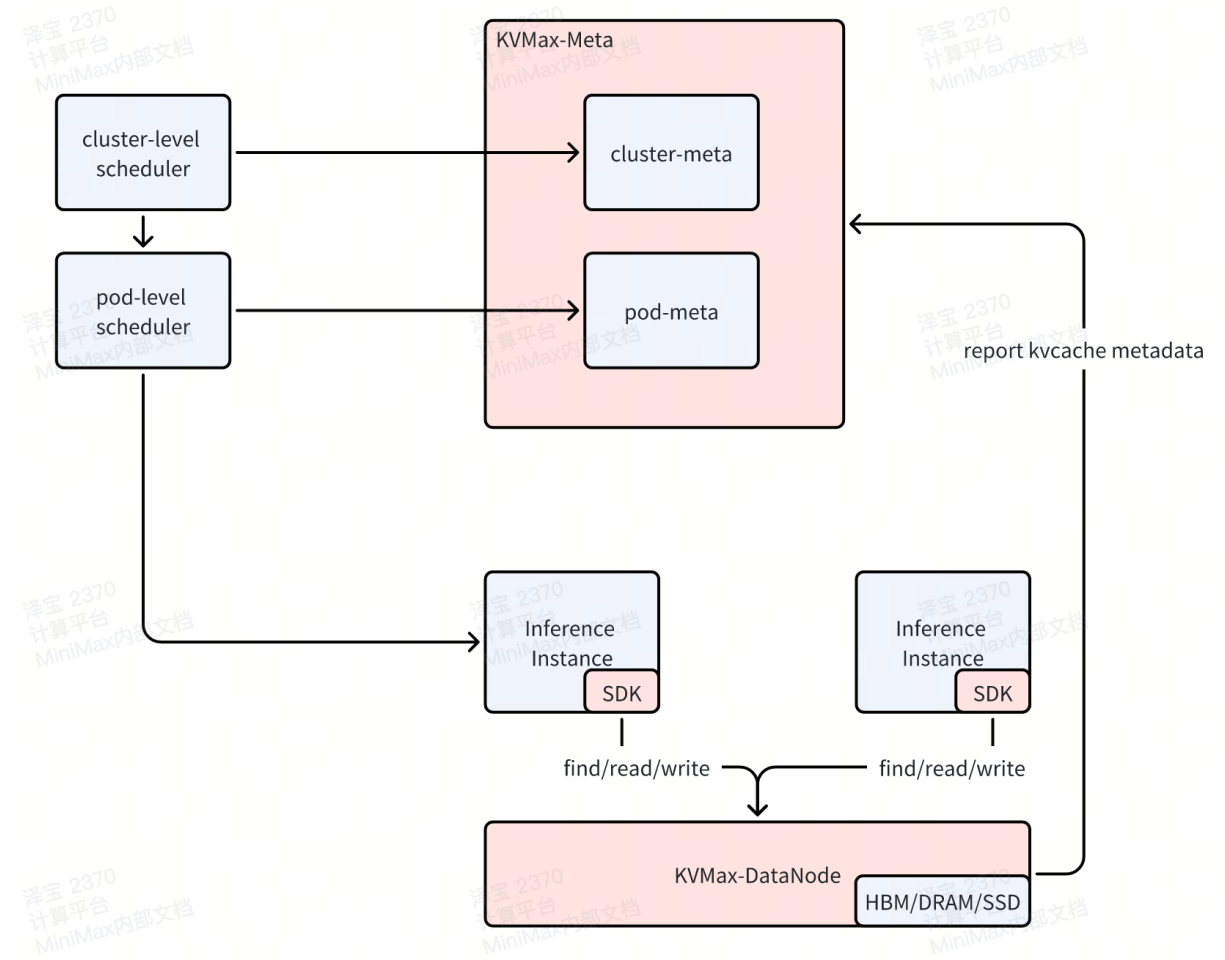
Challenges & Solutions

- Challenge 3: Sensitive to cache miss & QPS burst
 - 5% hit rate drop leads to 25% increase of prefill load
 - $(1 - 75\%) / (1 - 80\%) = 125\%$
 - 5% qps burst(uncached request) leads to 25% increase of prefill load:
 - $(1 * 20\% + 5\%) / (1 - 80\%) = 125\%$
 - especially openclaw-like traffic, burst at top-of-the-hour
- Solution:
 - precise kvcache-aware routing
 - high SLA of kvcache components
 - more buffer for burst

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KVMax: KVCache Mngmt at MiniMax

- DataPlane
 - Node-level Daemon Management
 - Radix Tree Indexing
 - Storage Hierarchy
- Control Plane
 - Two-Tier Centralized Meta
 - Scheduler Colocation
 - Local Daemon reports to Meta service



Comparisons with other KVCache mngmnt products

- Sames
 - Control-plane & Data plane Separation
- Diffs
 - simpler design
 - meta-service & scheduler colocation: multi-cluster

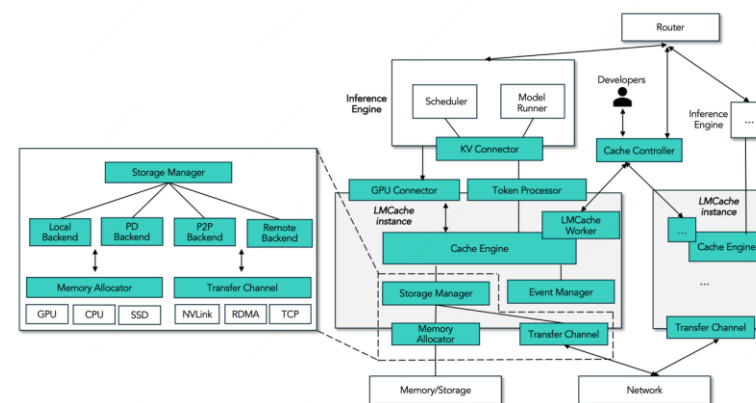
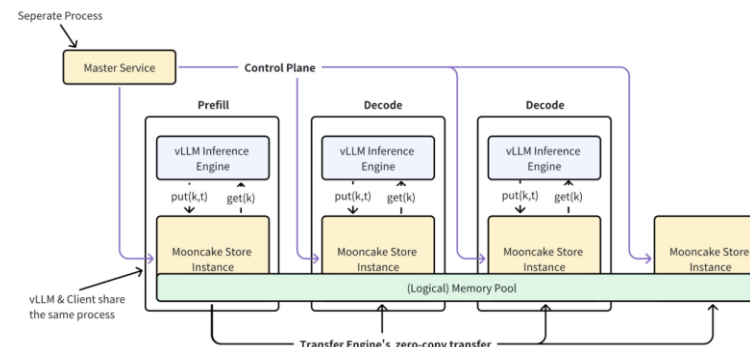


Figure 6: End-to-end system workflow for LMCACHE.

Rethink KVCache for Agents

- Insight-1: Rethink Cache Hierarchy: is block storage the answer?
 - Low Bandwidth: 6s of read/write cycle exceed the SSD bandwidth easily at low RPM
 - Low Effective Capacity: only (ssd bandwidth * 6s) capacity is effective, about 3%
 - Low Durability: high-frequency writes leads to rapid wear-out
 - What we need: high bandwidth, medium capacity storage
- Insight-2: Beyond Simple Cache-Awareness
 - Hot spot problem: pure kvcache-aware routing often leads to load imbalance. Mapping everything to the "best match" prefix can saturate a Pod's PCIe bandwidth, even if the hit rate is high.
 - The Missing Link: Schedulers currently cannot predict how many times a prefix will be reused in the future.
 - Potential Solutions:
 - Proactive Prediction: Estimating future hits when a request arrives.
 - Reactive Migration: Dynamically migrating KVCache once a hotspot is detected.

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Thanks for listening!



The banner features the MINIMAX logo in the top left, with a red and orange waveform icon. To its right is a red pill-shaped button containing the text 'M2.5' and a globe icon. The main title 'Coding Plan' is in large, bold black font, with 'MiniMax-M2.5' below it. A QR code is on the left, with the text 'Coding Plan 12% OFF!' underneath. A description states: 'A comprehensively upgraded general-purpose model with stronger reasoning, broader knowledge.' Below this is a URL in a red-bordered box: <https://platform.minimax.io/subscribe/coding-plan>. On the right, there is a cartoon character wearing a white sailor hat and holding a telescope, and a magnifying glass with a red handle focusing on a waveform icon.

MINIMAX M2.5

Coding Plan

MiniMax-M2.5

A comprehensively upgraded general-purpose model with stronger reasoning, broader knowledge.

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Coding Plan
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